Full dataset PCA

2024-11-06

Load gt

```
mad_bor_g1000_sum <- gt_load(path_resrepo("data/intermediate/rbind_data/mad_bor_g1000_sum.gt"))
```

Impute

```
mad_bor_g1000_sum <- gt_impute_simple(mad_bor_g1000_sum, method = "random")
gt_set_imputed(mad_bor_g1000_sum, TRUE)
```

LD clumping

```
to_keep_LD <- loci_ld_clump(mad_bor_g1000_sum, thr_r2 = 0.5, size = 50, return_id = TRUE)

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 1377
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 223798
```

Compute PCA

```
random_pca <- gt_pca_randomSVD(mad_bor_g1000_sum)
pcs <- as.data.frame(random_pca$u)
variance <- tidy(random_pca)
```

Add population metadata

```
pop_names <- read.csv(path_resrepo("data/raw/data_input/igsr-1000-genomes-phase-3-release.tsv"), sep = 

# find matching id in pop_names Sample.name and mad_bor_g1000_sum id, and for the matches, replace the 
for (i in seq_len(nrow(pop_names))) {
  if (pop_names$Sample.name[i] %in% mad_bor_g1000_sum$id) {
    mad_bor_g1000_sum[mad_bor_g1000_sum$id == pop_names$Sample.name[i], "population"] <- pop_names$Popul
```

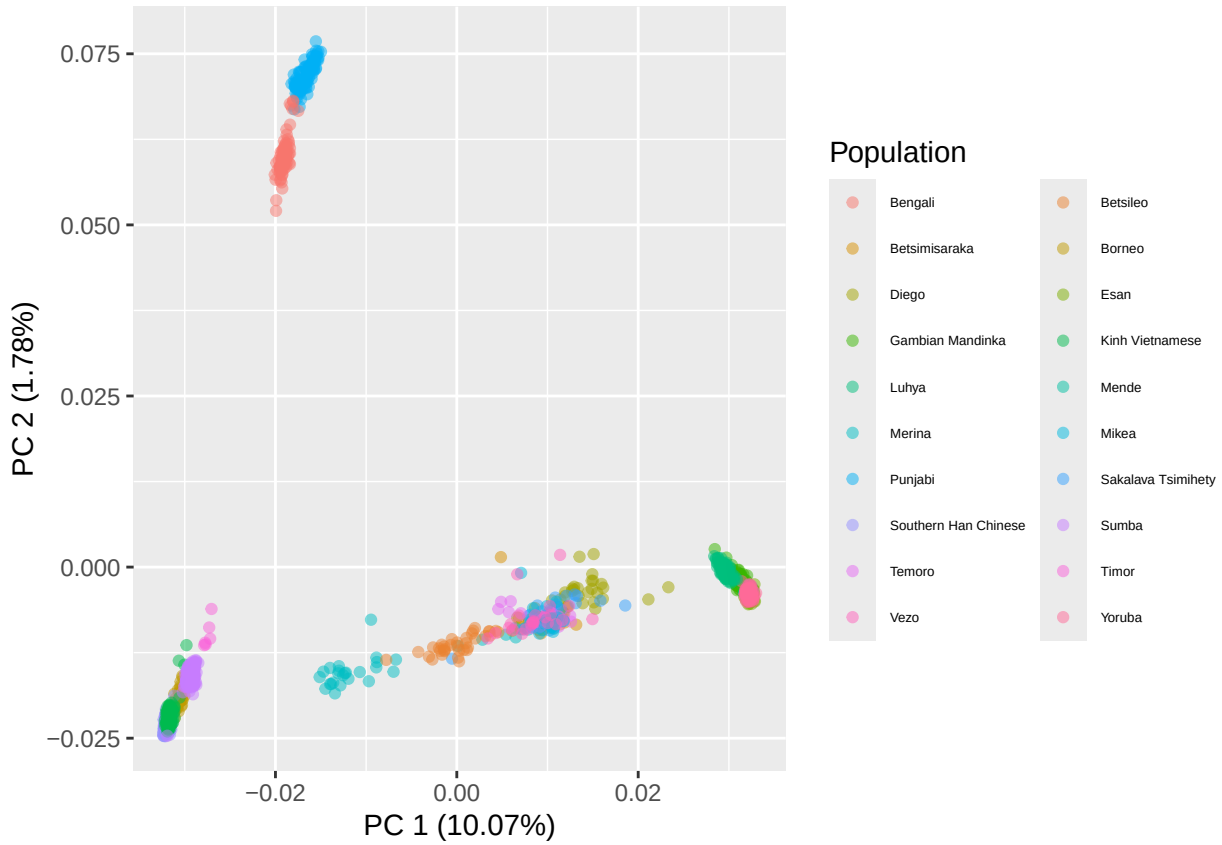
```
  }
}

# replace "Luhya,Luhya" with "Luhya"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Luhya,Luhya"] <- "Luhya"
# replace "Esan,Esan" with "Esan"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Esan,Esan"] <- "Esan"
# replace "Gambian Mandinka,Gambian" with "Gambian Mandinka"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Gambian Mandinka,Gambian"] <- "Gambian Ma
# replace "Mende,Mende" with "Mende"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Mende,Mende"] <- "Mende"
# replace "Kinh,Kinh Vietnamese" with "Kinh Vietnamese"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Kinh,Kinh Vietnamese"] <- "Kinh Vietnames
# replcae "Punjabi,Punjabi" with "Punjabi"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Punjabi,Punjabi"] <- "Punjabi"
# replcae "Benga,Bengali" with "Bengali"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Bengali,Bengali"] <- "Bengali"

# replace any _ with a space
mad_bor_g1000_sum$population <- gsub("_", " ", mad_bor_g1000_sum$population)
```

Plot PCA

```
ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$population)) +
  geom_point(alpha = 0.5) +
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", ro
  theme(legend.text = element_text(size = 5)) +
  guides(colour = guide_legend(ncol = 2, byrow = TRUE))
```



This PCA uses the KING relatedness filtering of 0.177, equivalent to a first degree relatives (parent, sibling, or child) filtered within each population.

Unbalanced sample sizes will cause bias in the dimensions of this first plot. To explore how this might be affecting the PCA, the next dataset will randomly subsample populations with more individuals down to equivalent sizes.

Create a dataset with broadly equal sample sizes

```
set.seed(1)

subsample <- function(x) {
  if (nrow(x) <= 30) {
    return(x)
  } else {
    sample_n(x, size = 30)
  }
}

mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  group_by(population) %>%
  group_modify(~ subsample(.x)) %>%
  ungroup()

mad_bor_g1000_sum %>%
```

```
  group_by(population) %>%
  count()
```

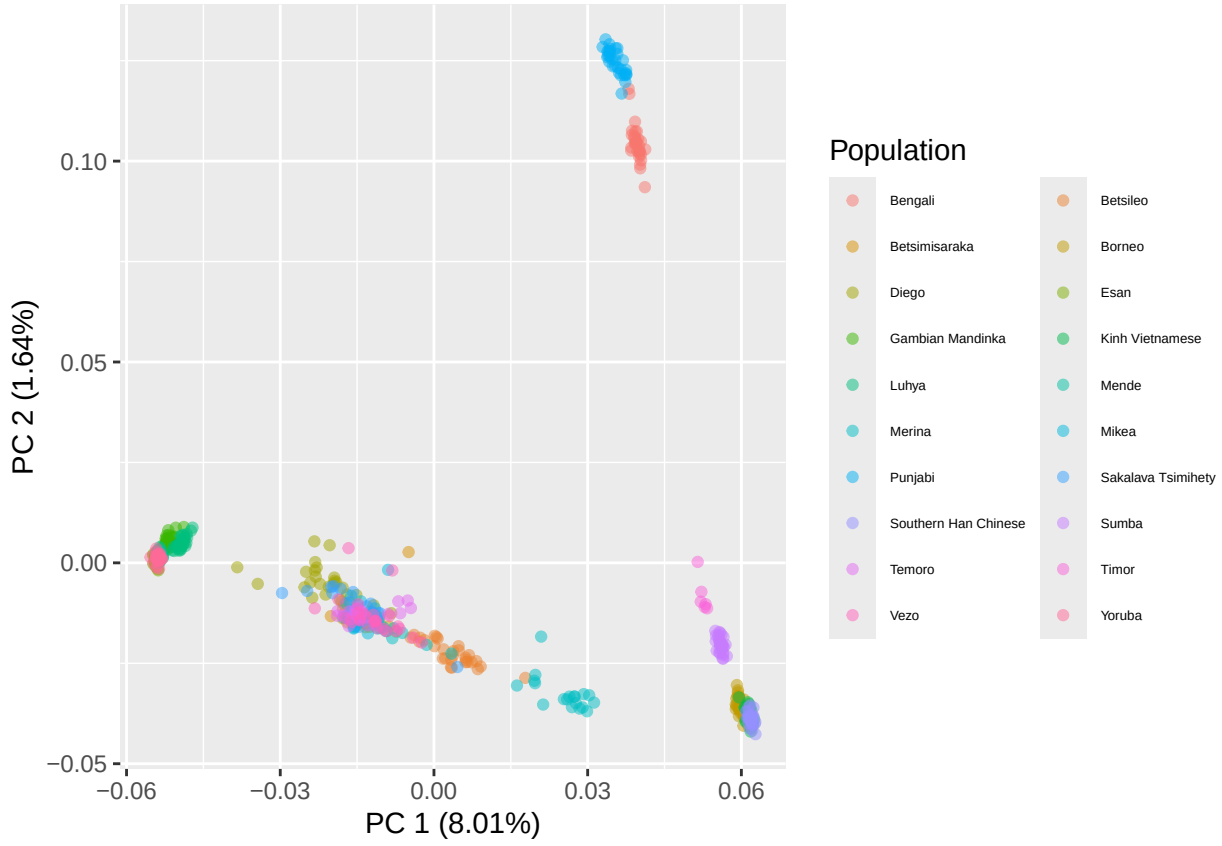
```
## # A tibble: 20 x 2
## # Groups:   population [20]
##    population               n
##    <chr>                <int>
##  1 Bengali                 30
##  2 Betsileo                30
##  3 Betsimisaraka           30
##  4 Borneo                  30
##  5 Diego                   30
##  6 Esan                    30
##  7 Gambian Mandinka        30
##  8 Kinh Vietnamese         30
##  9 Luhya                   30
## 10 Mende                   30
## 11 Merina                  30
## 12 Mikea                   18
## 13 Punjabi                 30
## 14 Sakalava Tsimihety      30
## 15 Southern Han Chinese    30
## 16 Sumba                   30
## 17 Temoro                  24
## 18 Timor                    6
## 19 Vezo                    24
## 20 Yoruba                  30
```

Compute PCA

```
sub_pca <- gt_pca_randomSVD(mad_bor_g1000_sum)
pcs <- as.data.frame(sub_pca$u)
variance <- tidy(sub_pca)
```

Plot PCA

```
ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$population)) +
  geom_point(alpha = 0.5) +
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", rou
  theme(legend.text = element_text(size = 5)) +
  guides(colour = guide_legend(ncol = 2, byrow = TRUE))
```



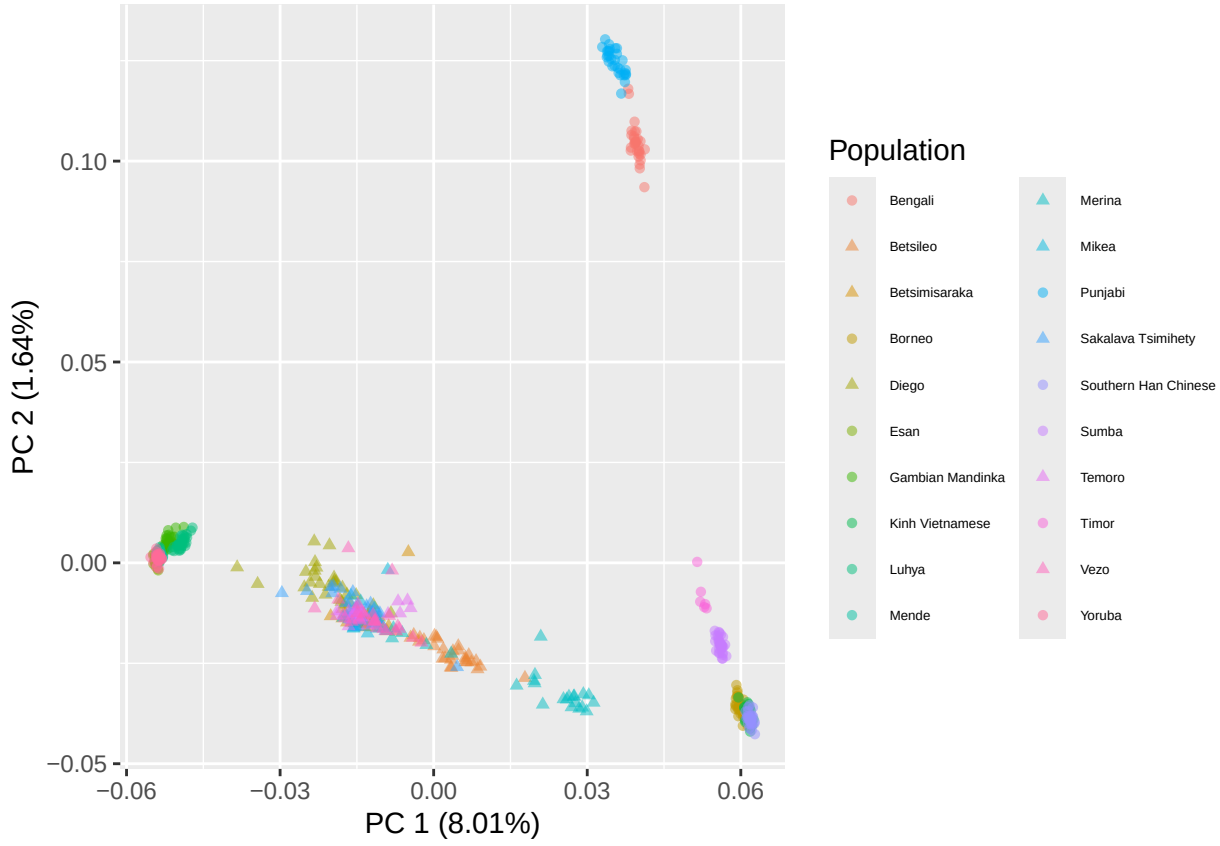
Altering the sample sizes reduces the explained variance of both principal components and increases the dispersion of the data.

Change legend

```
# create a new factor column with Malagasy populations as 1 and others as 0
mad_bor_g1000_sum$malagasy <- 0
all_malagasy <- subset(mad_bor_g1000_sum, population %in% c("Betsimisaraka", "Merina", "Diego", "Sakala
mad_bor_g1000_sum[mad_bor_g1000_sum$id %in% all_malagasy$id, "malagasy"] <- 1
```

Plot PCA

```
ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$population, shape = as.factor(mad_bor_g1000_
  geom_point(alpha = 0.5) + #
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", ro
  theme(legend.text = element_text(size = 5)) +
  guides(shape = "none", colour = guide_legend(override.aes = list(shape = c(16, 17, 17, 16, 17, 16, 16
```



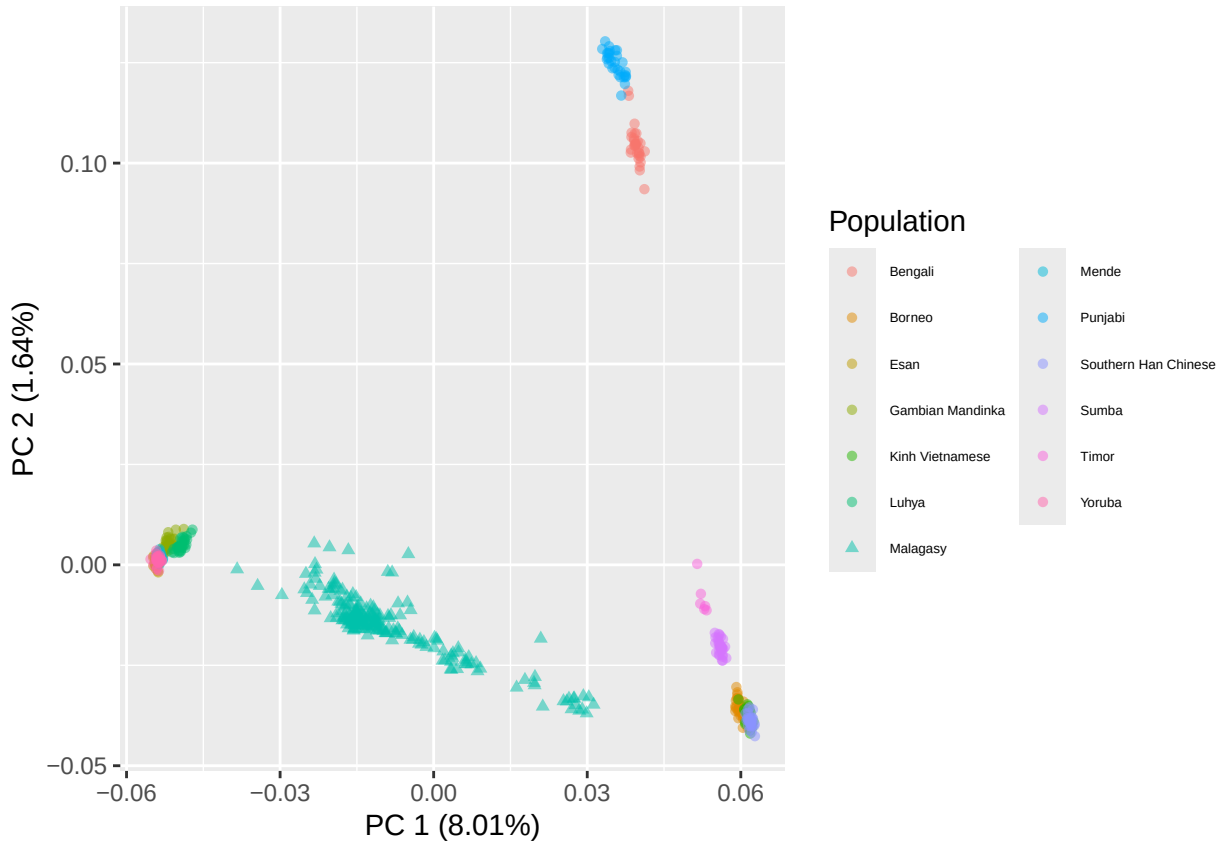

```
ggsave(paste0(res_dir, "pca_subsampled_malagasy_separated.jpg"), width = 7, height = 7, dpi = 300)
```

```
# create a new factor column with Malagasy populations as 1 and others as 0
mad_bor_g1000_sum$mad_vs_global <- "Other"

fill <- function(x) {
  if (x["malagasy"] == 1) {
    x["mad_vs_global"] <- "Malagasy"
  } else {
    x["mad_vs_global"] <- x["population"]
  }
}

mad_bor_g1000_sum$mad_vs_global <- apply(mad_bor_g1000_sum, 1, FUN = fill)

ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$mad_vs_global, shape = as.factor(mad_bor_g10
  geom_point(alpha = 0.5) + #
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", rou
  theme(legend.text = element_text(size = 5)) +
  guides(shape = "none", colour = guide_legend(override.aes = list(shape = c(16, 16, 16, 16, 16, 16, 17
```

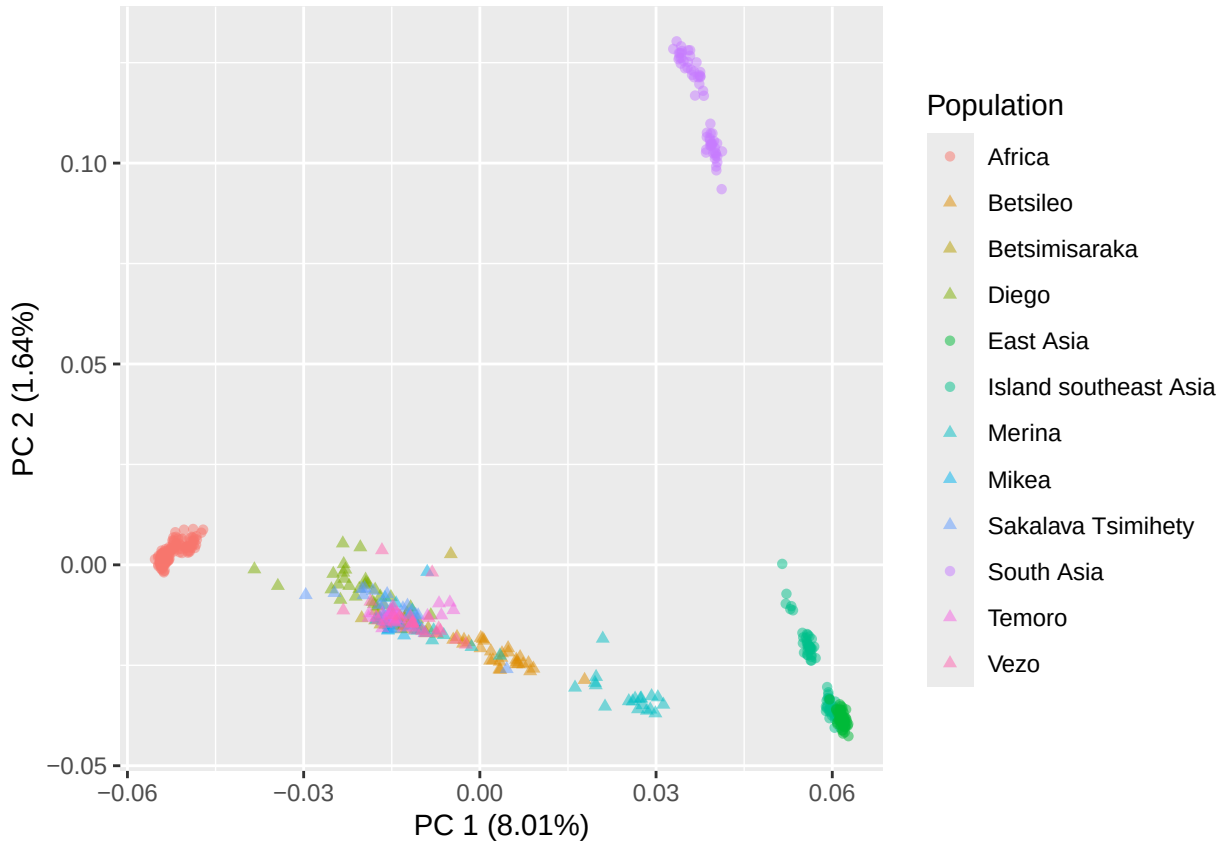



```
ggsave(paste0(res_dir, "pca_subsampled_malagasy_together.jpg"), width = 7, height = 7, dpi = 300)
```

```
mad_bor_g1000_sum$Region <- "Other"

mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Bengali", "Punjabi"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Southern Han Chinese", "Kinh Vietnamese"), "Regi
# Not really happy about grouping all African samples here though...
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Esan", "Gambian Mandinka", "Luhya", "Mende", "Yo
mad_bor_g1000_sum[mad_bor_g1000_sum$malagasy == 1, "Region"] <- mad_bor_g1000_sum[mad_bor_g1000_sum$mala

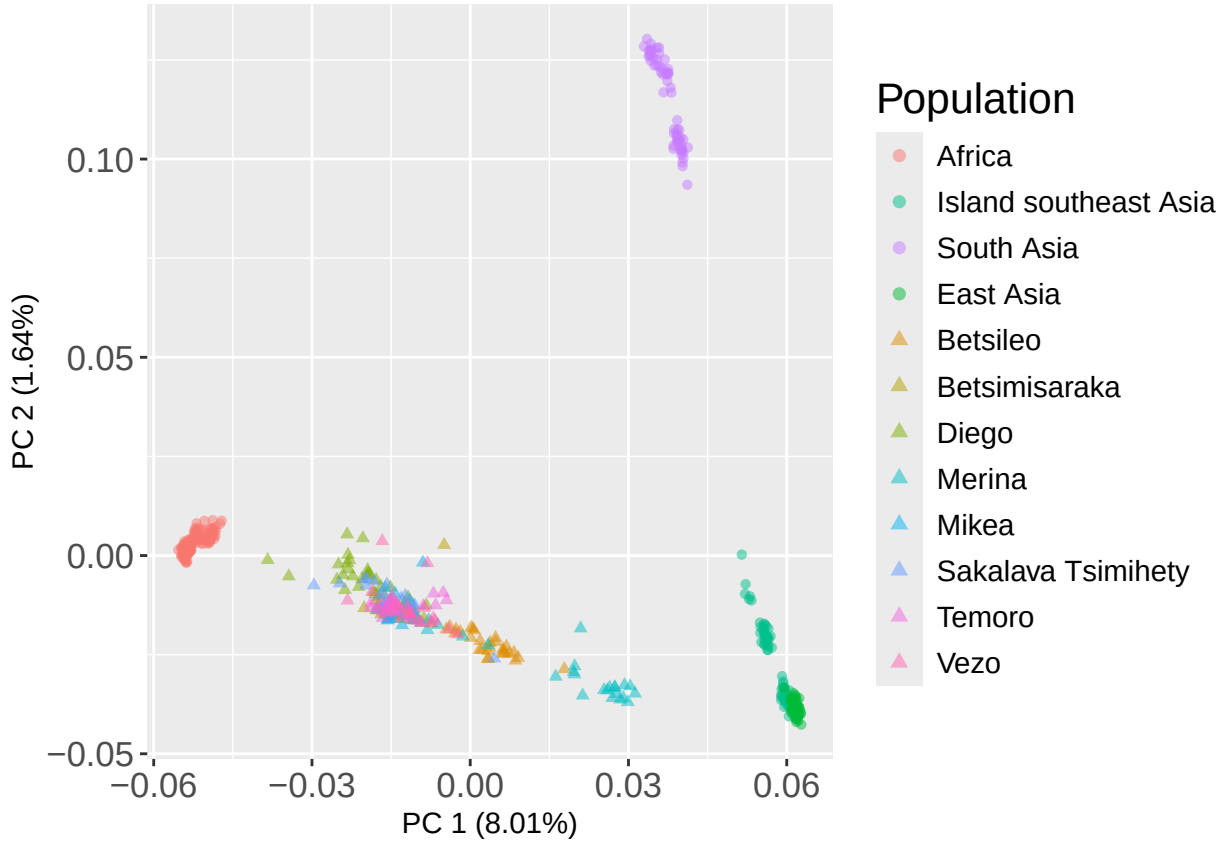

ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$Region, shape = as.factor(mad_bor_g1000_sum$r
  geom_point(alpha = 0.5) +
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", rou
  theme(legend.position = "right") +
  guides(shape = "none", colour = guide_legend(override.aes = list(shape = c(16, 17, 17, 17, 16, 16, 17
```




```
ggsave(paste0(res_dir, "pca_subsampled_regions_together.jpg"), width = 7, height = 7, dpi = 300)
```

```
mad_bor_g1000_sum$Region <- factor(mad_bor_g1000_sum$Region, levels = c("Africa", "Island southeast Asia

ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$Region, shape = as.factor(mad_bor_g1000_sum$r
  geom_point(alpha = 0.5) +
  labs(title = NULL, x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"), y = paste0("PC 2 (", rou
  theme(legend.position = "right") +
  guides(shape = "none", colour = guide_legend(override.aes = list(shape = c(16, 16, 16, 16, 17, 17, 17
  theme(plot.title = element_text(size = 16), axis.text = element_text(size = 13), legend.title = eleme
  scale_colour_manual(
    values = c(
      "Africa" = "#F8766D",
      "East Asia" = "#00BA38",
      "South Asia" = "#C77CFF",
      "Island southeast Asia" = "#00C08B",
      "Betsileo" = "#DE8C00",
      "Betsimisaraka" = "#B79F00",
      "Diego" = "#7CAE00",
      "Merina" = "#00BFC4",
      "Mikea" = "#00B4F0",
      "Sakalava Tsimihety" = "#619CFF",
      "Temoro" = "#F564E3",
      "Vezo" = "#FF64B0"
    )
  )
```




```
ggsave(paste0(res_dir, "pca_subsampled_regions_together_large_legend.jpg"), width = 7, height = 7, dpi =
```